

3(a)	extension is proportional to (applied) force	B1
3(b)(i)	P at (60, 5.4)	A1
3(b)(ii)	E at (80, 5.9)	A1
3(c)(i)	$k = F/x$ or $k = \text{gradient of (straight line section of) graph}$	C1
	e.g. gradient = $5.4 / 0.060$	A1
	$k = 90 \text{ N m}^{-1}$	
3(c)(ii)	Young modulus or $E = \sigma/\epsilon$ or FL/Ax or kL/A	C1
	$E = (5.4 \times 3.2) / (4.0 \times 10^{-7} \times 0.06)$ or $90 \times 3.2 / (4.0 \times 10^{-7})$	C1
	$E = 7.2 \times 10^8 \text{ Pa}$	A1
3(d)	work done = area under graph	B1
	= $(1.0 \pm 0.2) \text{ J}$	A1
3(e)	the extension will be smaller (for the same force on the thicker sample) or a greater force is required (to extend the thicker sample by the same amount) or spring constant is proportional to area	M1
	the spring constant (of the second sample) will be greater	A1